

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TELEMATIC CONTROL MODULE (TCU)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Telematic Control Module (TCU). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Information and Entertainment System](#)

(415-01 Information and Entertainment System, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B10A2-12	Crash Input - Circuit short to battery	Crash signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to power
B10A2-14	Crash Input - Circuit short to ground or open	Crash signal circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, open circuit, high resistance
B10A2-38	Crash Input - Signal frequency incorrect	- Crash signal circuit short circuit to another signal circuit - Supplementary restraint system fault	- Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to another signal circuit - Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
B116D-11	SOS On Call Keypad - Circuit short to ground	NOTE: Circuit reference - ECALL + BCALL BTN Telematic switchpack circuit short circuit to ground	Refer to the electrical circuit diagrams and check the telematic switchpack circuit for short circuit to ground
B116D-15	SOS On Call Keypad - Circuit short to battery or open	NOTE: Circuit reference - ECALL + BCALL BTN	Refer to the electrical circuit diagrams and check the telematic switchpack circuit for short circuit to power, open circuit, high resistance

		Telematic switchpack circuit short circuit to power, open circuit, high resistance	
B116D-97	SOS On Call Keypad - Component or system operation obstructed or blocked	NOTE: Circuit reference - ECALL + BCALL BTN - Telematic switchpack circuit short circuit to ground, short circuit to power, open circuit, high resistance - Telematic switchpack switch stuck active	NOTE: Switch signals are blocked following false or repeated operation. - Refer to the electrical circuit diagrams and check the telematic switchpack circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Test the operation of the telematic switchpack switches. Install a new telematic switchpack as necessary
B116E-11	SOS On Call Illumination - Circuit short to ground	NOTE: Circuit reference - BCALL ACTIVATED LED / BCALL BTN BACKLIGHT / ECALL ACTIVATED LED / ECALL BTN BACKLIGHT Telematic switchpack illumination circuit short circuit to ground	Refer to the electrical circuit diagrams and check the telematic switchpack illumination circuit for short circuit to ground
B116E-15	SOS On Call Illumination - Circuit short to	Telematic switchpack illumination circuit short circuit to	Refer to the electrical circuit diagrams and check the telematic switchpack

	battery or open	power, open circuit, high resistance	illumination circuit for short circuit to power, open circuit, high resistance
B116F-09	Telematics SIM Card - Component failures	Telematics control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new telematics control module
B1179-04	Integrated GSM/GPRS Unit - System internal failures	Telematics control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new telematics control module
B117A-12	Backup Microphone - Circuit short to battery	Microphone circuit short circuit to power	Refer to the electrical circuit diagrams and check the microphone circuit for short circuit to power
B117A-14	Backup Microphone - Circuit short to ground or open	Microphone circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the microphone circuit for short circuit to ground, open circuit, high resistance
B119F-01	GPS Antenna - General electrical failure	Global positioning system antenna circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the global positioning system antenna circuit for short circuit to ground, short circuit to power
B119F-13	GPS Antenna - Circuit open	Global positioning system antenna circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the global positioning system antenna circuit for open circuit, high resistance
B1252-01	USB Port - General electrical failure	- USB interface circuit short circuit to ground, short circuit to power - High voltage applied to USB interface	- Refer to the electrical circuit diagrams and check the USB interface circuit for short circuit to ground, short circuit to power Repair circuit as required. Clear the DTCs and retest - Check USB device

B1252-13	USB Port - Circuit open	▪ USB device circuit open circuit, high resistance ▪ USB interface circuit open circuit, high resistance ▪ No USB device connected	▪ Refer to the electrical circuit diagrams and check the USB interface circuit for open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ Check USB device
B1252-87	USB Port - Missing message	▪ Unable to establish internet connection using customer SIM ▪ Communication with SIM failing ▪ 3G WIFI module internal failure	▪ Check operation of customer SIM ▪ Clear the DTCs and retest. If the fault persists, install a new 3G WIFI module
B1A56-11	Antenna - Circuit short to ground	▪ GSM antenna circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the GSM antenna circuit for short circuit to ground
B1A56-15	Antenna - Circuit short to battery or open	▪ GSM antenna circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the GSM antenna circuit for short circuit to power, open circuit, high resistance
B1A01-12	Speaker #1 - Circuit short to battery	▪ Telematics speaker circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the telematics speaker circuit for short circuit to power
B1A01-14	Speaker #1 - Circuit short to ground or open	▪ Telematics speaker circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the telematics speaker circuit for short circuit to ground, open circuit, high resistance
B1D98-01	Microphone Audio Output - General electrical failure	▪ Microphone output circuit short circuit or open circuit	▪ Refer to the electrical circuit diagrams and check the microphone audio output circuit between the telematics control unit and the integrated audio

			module for short circuit to ground or power, open circuit
P0560-04	System Voltage - System internal failures	▪ Telematics control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the electrical circuit diagrams and check the telematics control module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Missing message from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0010-87	Medium Speed CAN Communication Bus - Missing message	▪ Missing message from another control module via the medium speed CAN bus (body)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0010-88	Medium Speed CAN Communication	▪ Medium speed CAN bus (body) circuit short circuit to	▪ Using the manufacturer approved diagnostic system, perform a CAN network

	Bus - Bus off	ground, short circuit to power, open circuit, high resistance	integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
U201A-51	Control Module Main Calibration Data - Not programmed	Telematics control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the telematics control module with the latest level software
U201A-54	Control Module Main Calibration Data - Missing calibration	Telematics control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the telematics control module with the latest level software
U2109-09	Back-up Battery - Component failures	- Backup battery poor connection or not connected - Telematics control module internal battery circuit open circuit, high resistance - Telematics control module internal battery failure	NOTE: Following an emergency call or stolen vehicle tracking alert, a new telematics control module internal battery must be installed before this DTC can be cleared - Check backup battery for poor connection or not connected - Refer to the electrical circuit diagrams and check the telematics control module internal battery circuit for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and perform

			routine - Test Internal Backup Battery (0x040C). If the fault persists, install a new telematics control module internal battery. Using the manufacturer approved diagnostic system, reset the non rechargeable battery power level following the backup battery replacement process
U2109-91	Back-up Battery - Parametric	▪ Low charge in backup battery ▪ Telematics control module internal battery failure	NOTE: Following an emergency call or stolen vehicle tracking alert, a new telematics control module internal battery must be installed before this DTC can be cleared ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Test Internal Backup Battery(0x040C) ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Test Internal Backup Battery (0x040C). If the fault persists, install a new telematics control module internal battery. Using the manufacturer approved diagnostic system, reset the non rechargeable battery power level following the backup battery replacement process
U3000-44	Control Module - Data memory failure	▪ Telematics control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new telematics control module
U3000-49	Control Module -	▪ Telematics control module internal real	▪ Using the manufacturer approved diagnostic system,

	Internal electronic failure	time clock or global position system communication failure	clear the DTCs and retest. If the fault persists, install a new telematics control module
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